



Beata Zalewa

Zadbaj o bezpieczeństwo aplikacji
w chmurze dzięki Microsoft Defender
for Cloud Apps



Azure Summit 2023

All about cloud technologies **ONLINE**

30.03.2023

Godz. 15:00

 Future Processing

lingaro

 directpl

Demant

mscloud

kursySQL.pl

Millennium
bank

PROMISE
GRUPA APN PROMISE

Trek2Summit

Etteplan

integrity
partners



Secure access for applications with Defender for Cloud Apps

Beata Zalewa

About



In professional life Lead Technical Architect, consultant, Microsoft Certified Trainer, developer, freelancer.



From the beginning of career associated with Microsoft technologies.



Private life in numbers: 1 husband, 1 daughter, 1 cat and 2 dogs. My hobbies are detective stories and photography.



Email: info@zalnet.pl



<https://www.linkedin.com/in/beatazalewa/>



WWW: <https://blog.zalnet.pl>

Solution overview

What is **Defender for Cloud Apps**?

A native *Cloud Access Security Broker (CASB)* that supports various deployment modes including log collection, API connectors, and reverse proxy.

Cloud app discovery platform that fetches data from Defender for Endpoint or Firewall logs

Granular control interface of collaboration and access to SaaS (Software as a service) apps in the web (security and compliance enablement)

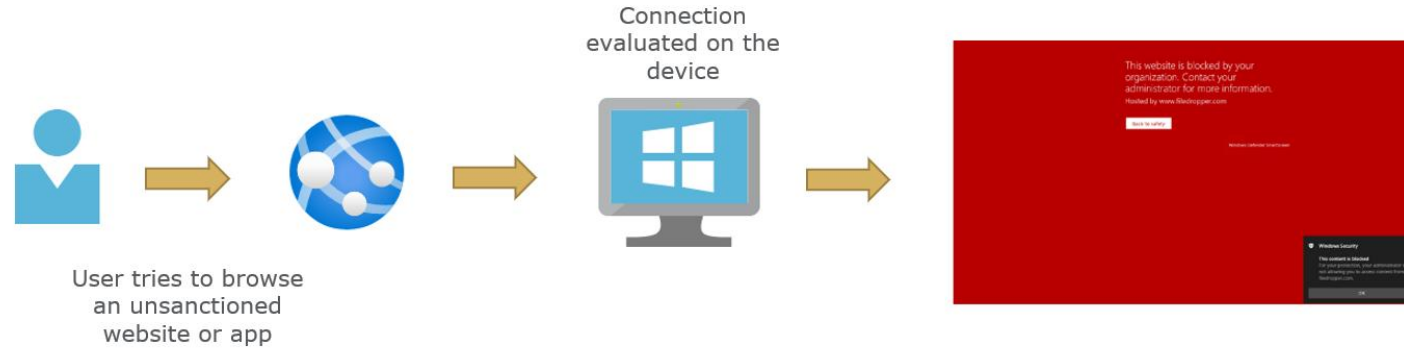
Decentralized / cloud-based solution where the security perimeter is built-in the Defender Suite

How the cloud changed the enterprise?

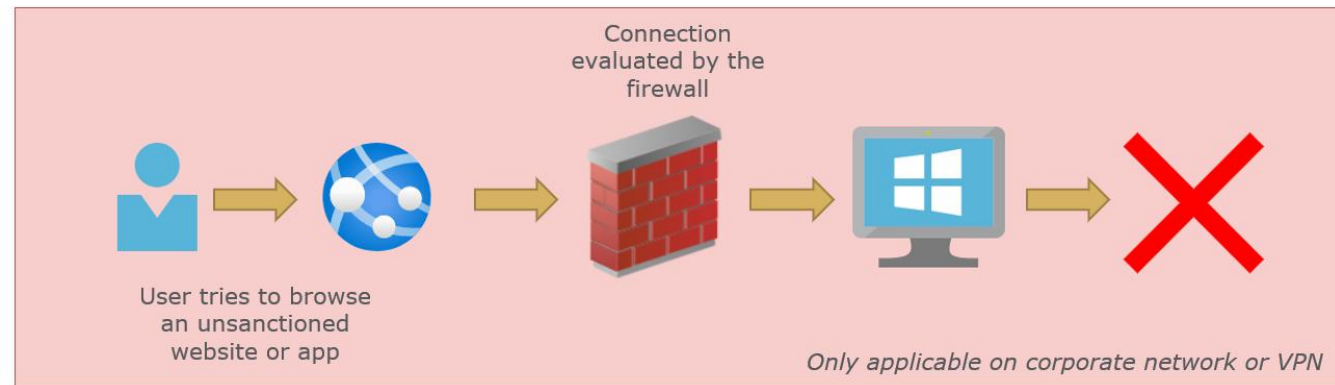
The modern vs. legacy approach on network controls differentiate through a central component, the firewall, which in the legacy flow was needed to control connections from a client.

The modern approach is that connections are evaluated on the device and thus independently of the network.

Modern



Legacy



How the cloud changed the enterprise?

License requirements

- Microsoft Cloud App Security Licensing Datasheet

[RE2NXYO \(microsoft.com\)](#)

- When talking about pricing, Defender for Cloud Apps has some included features in Azure AD Premium P1, that is part of Enterprise Mobility + Security E3.

- The full capabilities get available with the Microsoft 365 E5 Security or the standalone CAS license, which costs ~ 3.5\$/user/month.

[https://aka.ms/M365EnterprisePlans](#)

What are Cloud Apps?

The challenge consists of thousands of Cloud applications and websites that users need for collaboration.

Shadow IT describes non IT-personnel, that interacts with separate accounts (no SSO or central IdP) on cloud infrastructure/apps that the organization utilizes.

The problem is, that the IT department has no governance over these actions and lacks of security and compliance.

Microsoft 365 Defender

Search

Cloud app catalog

Filters:

Apps: App tag: ☒ Sanctioned ☐ Unsanctioned ☐ None Risk score: 0 10

Compliance risk factor: **Select factors** Security risk factor: **Select factors**

Browse by category:

Hosting services	3.5K
IT services	3.1K
Accounting and finance	2.3K
Business management	2K
Productivity	1.7K
Human-resource manage...	1.2K
E-commerce	1.2K
Marketing	1.2K
Education	1.1K

☐ Bulk selection New policy from search

App	Risk score	Actions
Microsoft Xbox News and entertainment	10	☒ ☐ ⋮
Xamarin Development tools	10	☒ ☐ ⋮
Azure Virtual Desktop Cloud computing platform	10	☒ ☐ ⋮
Microsoft Word Online Productivity	10	☒ ☐ ⋮
Microsoft Azure Services Cloud computing platform	10	☒ ☐ ⋮
Windows365 Cloud computing platform	10	☒ ☐ ⋮

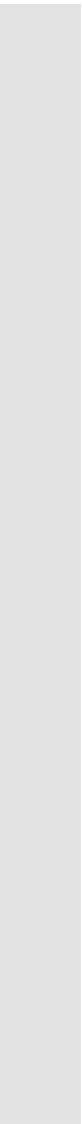
Cloud App catalog

Microsoft's own database of discovered apps in the web has over 30'000 entries. They are added on the go.

Before
demo



**KEEP
CALM
AND
PRAY THE DEMO
WORKS**



DEMO 1

**Cloud app
catalog**

App details

Each app (example LinkedIn) has a lot of information assigned, such as general information, security related data and governance standards. As you can see, there is a lot of useful information that an ordinary consumer might never have known.

The screenshot displays the LinkedIn app details page. At the top, there is a search bar and a navigation menu with 'Education' selected. The main content area is divided into three sections: General, Security, and Compliance. Each section has a score of 10 out of 10, indicated by a green bar. The General section lists various company details, the Security section lists various security features, and the Compliance section lists various compliance standards.

Education 1

Search

LinkedIn Social network

LinkedIn is a professional networking site that allows its members to create business connections, search jobs, and find potential clients. [Sug](#)

GENERAL 10

Category: So...	Headquarters: Uni...	Data center: Unite...
Founded: 19...	Holding: Public	Domain: 2 *.linke...
Domain registration:	Consumer popularity:	Privacy policy: priv...
Vendor: Micr...	Data types: 2 Do...	✓ Disaster Recovery...

SECURITY 10

Latest breach:	Data-at-rest encryption method: ✓ Multi-factor auth...	
✓ User audit t...	✓ Admin audit trail	✓ Data audit trail
✓ Data classifi...	✓ Remember passw...	✓ User-roles support
✓ Valid certific...	✓ Trusted certificate	Encryption protocol:
✓ HTTP securi...	✓ Supports SAML	✓ Protected against...
✓ Requires us...	✓ Password policy	

COMPLIANCE 10

✓ ISO 27001	✓ ISO 27018	✓ ISO 27017
✓ FINRA	✓ FISMA	✓ GAAP

How to make it work?



You may ask yourself what to do with this data.



Usually in later step policies are enforced to strictly define which attributes of a cloud app are relevant or even required before accessing an app.



Score metrics can be configured to individual preferences and importance's.

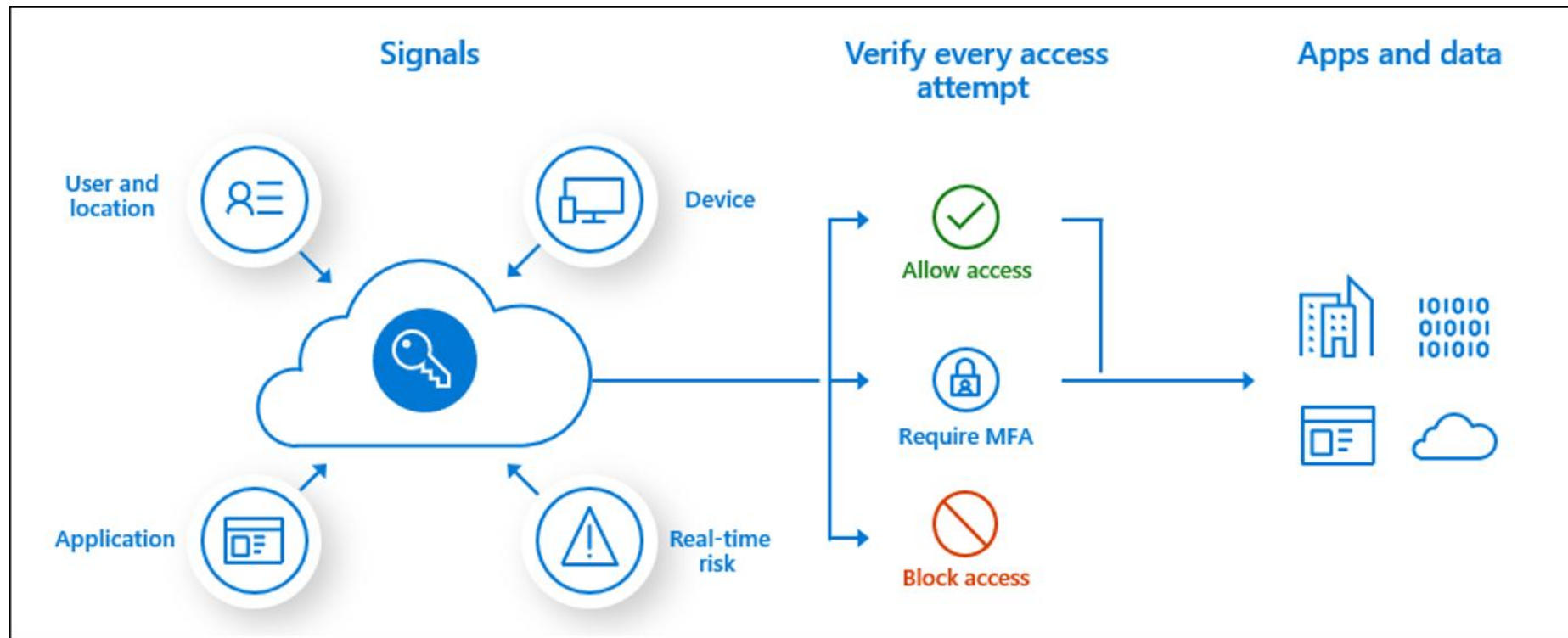


Furthermore, session controls work with **Conditional Access**.

Session control

Through conditional access, we also have the great opportunity to take control even at the session level.

- **Use app enforced restrictions** this only works with Sharepoint Online and Exchange Online and can create a limited experience within the apps.
- **Use conditional access app control** uses Microsoft Cloud App security where you can protect data with Conditional Access App Control by applying access and session controls.
- **Sign-in frequency** defines the time period before a user is asked to sign in again when attempting to access a resource. You can set this from a few hours up to 365 days.
- A **persistent browser session** allows users to remain signed in after closing and reopening their browser window. The option "stay signed-in" will also be skipped.



Conditional access

App tags

Apps can be tagged as:

Sanctioned - app is generally allowed and accessible

Unsanctioned - app is generally not allowed and might be not accessible

Monitored - app is under special observation, user receives a corresponding notification, that the access to this app is monitored

Individual tag - individual, for utilization in own policies

Microsoft 365 Defender

Search

Cloud app catalog

Filters:

Apps: App tag: ☒ Sanctioned ☐ Unsanctioned ☐ None Risk score: 0 10

Compliance risk factor: **Select factors** Security risk factor: **Select factors**

Advanced filters

Browse by category:

Search for category

- Social network 2
- Education 1
- Marketing 1

Bulk selection ☐ New policy from search ☐

App	Risk score	Actions
LinkedIn marketing solutions Marketing	10	☒ ☐ ☐
LinkedIn Learning Education	10	☒ ☐ ☐
LinkedIn Social network	10	☒ ☐ ☐
LinkedIn Engineering Social network	10	☒ ☐ ☐

Reports

Audit

App tags



DEMO 2

App tags



Each cloud app is rated by Microsoft by an individual risk score to:



Scores from **0** (worst, harmful) up to **10** (best, harmless)



App scores can be used to identify unwanted or malicious apps

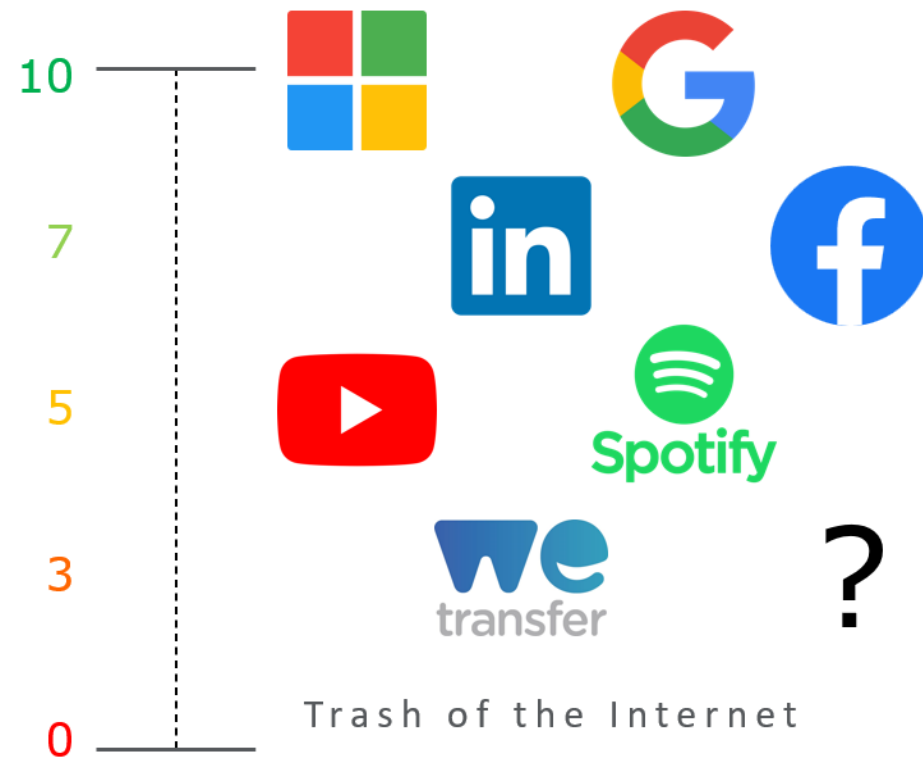


Score metrics can be configured to own concerns, but Microsoft raises data and provides default score metrics

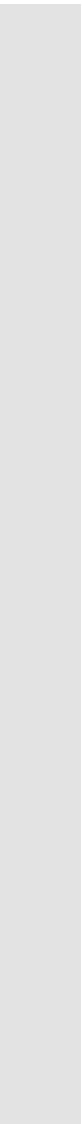


Block access to apps under a certain score (with Defender for Endpoint integration)

App risks score



App risks score



DEMO 3

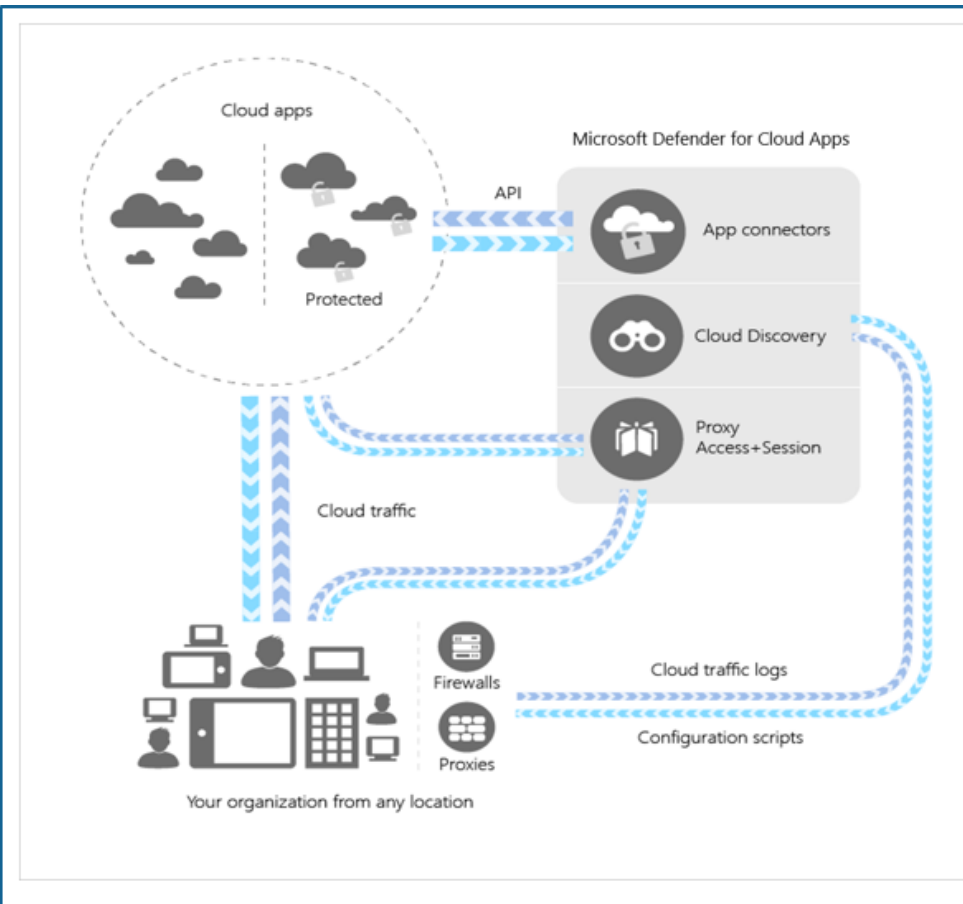
**App risks
score**

The Defender for Cloud Apps framework provides the following threat intelligence protection:

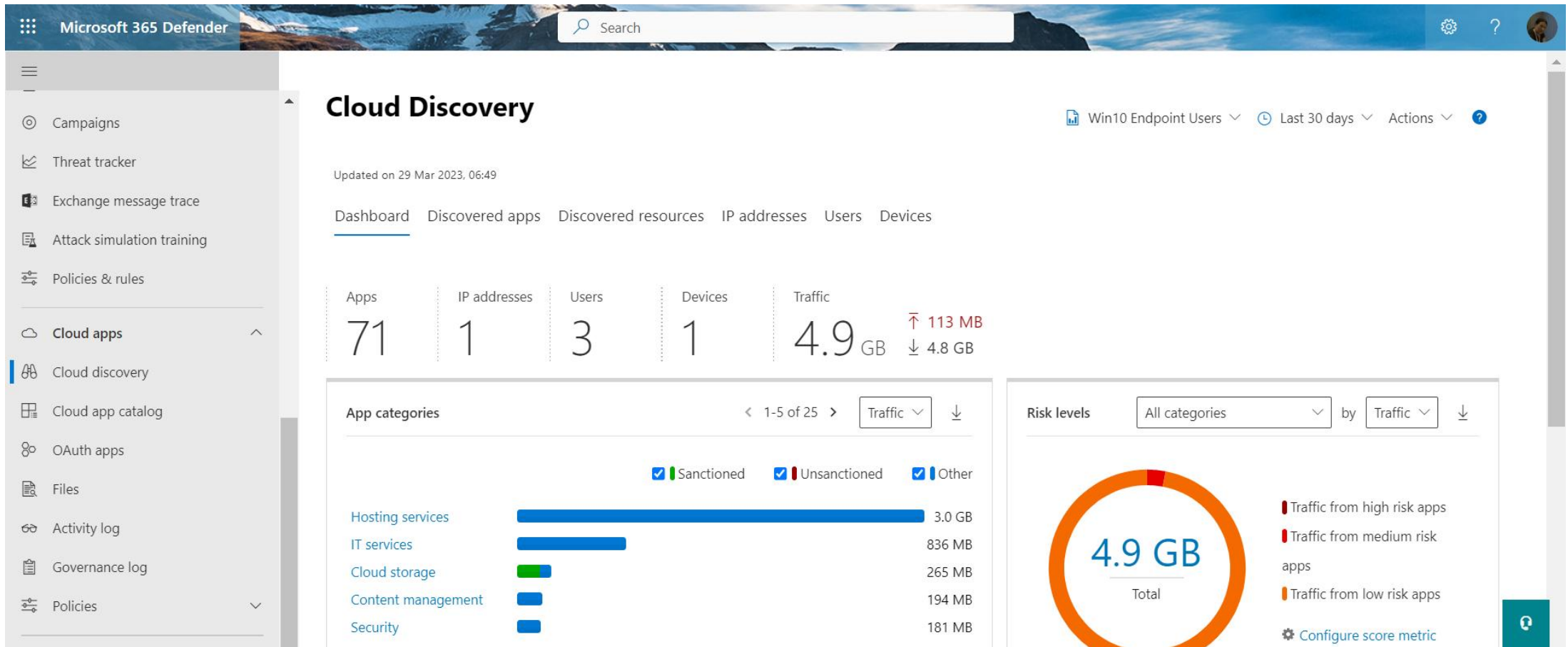
- Discover and control the use of Shadow IT
- Protect your sensitive information anywhere in the cloud
- Protect against cyberthreats and anomalies
- Assess the compliance of your cloud apps

Microsoft Defender for Cloud Apps architecture enables:

- Cloud Discovery
- Sanctioning and unsanctioning an app
- App connectors
- Conditional Access App Control protection
- Policy control



Explore Microsoft Defender for Cloud Apps



Cloud discovery

Microsoft 365 Defender

Search

Cloud Discovery

Win10 Endpoint Users Last 30 days Actions

Updated on 29 Mar 2023, 06:49

Dashboard Discovered apps Discovered resources IP addresses Users Devices

Queries: Select a query Save as Advanced filters

Apps: Apps... App tag: Sanctioned Unsanctioned None Risk score: 0 4

Compliance risk factor: Select factors Security risk factor: Select factors

Browse by category:

Search for category

Advertising 2

Data analytics 1

1 - 3 of 3 discovered apps

	Risk score	Tags	Traf...	Upl...	Tran...	Users	IP a...	Devices	Last...	Actions
☒	4		50 KB	—	1	1	1	1	28 Ma...	✓ ✕ ⋮
☐	4		26 KB	—	1	1	1	1	28 Ma...	✓ ✕ ⋮
☐	4		972 KB	39 KB	3	1	1	1	28 Ma...	✓ ✕ ⋮

Cloud discovery

Microsoft 365 Defender

Search

Campaigns

Threat tracker

Exchange message trace

Attack simulation training

Policies & rules

Cloud apps

Cloud discovery

Cloud app catalog

OAuth apps

Files

Activity log

Governance log

Policies

Policy management

Policy templates

Reports

Activity log

Investigate 6 months back

To improve performance, the Activity Log now displays activities from the last 30 days. It is possible to display more than 30 days of activities in the Investigate the last 6 months option. [Learn more](#)

Queries: Select a query

Save as

Advanced filters

App: Select apps

User name: Select users

Raw IP address: Enter IP address

Activity type: Select value

Location: Select countries/regions

+ New policy from search

Export

1 - 20 of 23 activities

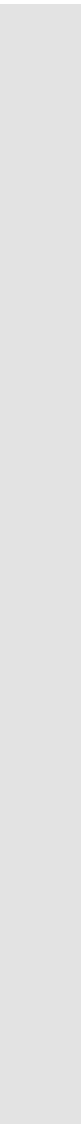
Show details

Hide filters

Table settings

Activity	User	App	IP address	Location	Device	Date
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		30 Mar 2023 10:01
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		30 Mar 2023 08:41
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		29 Mar 2023 20:20
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		29 Mar 2023 19:00
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		11 Mar 2023 08:25
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		11 Mar 2023 06:21
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		10 Mar 2023 22:51
Log on	Beata Zalewa	Microsoft De...	109.243.142.230	Poland		10 Mar 2023 21:30

Discover and investigate



DEMO 4

**Cloud
discovery**



Policies and templates

Policy and templates are used to create alert definition and generation. And this is really the key point, why we are doing this.

Difference between a policy and a template:

Policies - are active templates that will produce alerts

Templates - are blueprints that can be used to create policies



Policies allow you to define the way you want your users to behave in the cloud



There are multiple types of policies that correlate to the different types of information you want to gather about your cloud environment and the types of remediation actions you may want to take



The Microsoft Defender for Cloud Apps engine combines three aspects under each policy:

- Content scan based on preset templates or custom expressions
- Context filters
- Automated actions for governance and remediation

Configure file policies in Microsoft Defender for Cloud Apps

Microsoft 365 Defender

Search

Cloud discovery

Cloud app catalog

OAuth apps

Files

Activity log

Governance log

Policies

Policy management

Policy templates

Reports

Audit

Health

Permissions

Settings

More resources

Azure AD identity protection policies have been removed from the Defender for Cloud Apps policy list. To configure alerts from these policies, [edit the global alert service settings](#)

The Alerts/SMS (text messages) has been deprecated. If you would like to receive text alerts, you should use Microsoft Power Automate for custom alert automation. For more information, see [Integrate with Microsoft Power Automate for custom alert automation](#).

Customize alerts and actions by creating policies: [Create policy](#)

Threat detection

Information protection

Conditional access

Shadow IT

All policies

Filters:

Name: Policy name

Type: Select type

Status: ACTIVE DISABLED

Severity: High

Category: Select risk category

Advanced filters

+ Create policy

Export

1 - 20 of 33 Policies

Hide filters

Table settings

Policy	Count	Severity	Category	Action	Modified
Suspicious inbox manipulation rule This policy profiles your environment and triggers alerts when suspic...	0 open alerts	High	Threat detection		20 Mar 2023
Ransomware activity This policy profiles your environment and triggers alerts when an act...	0 open alerts	High	Threat detection		20 Mar 2023

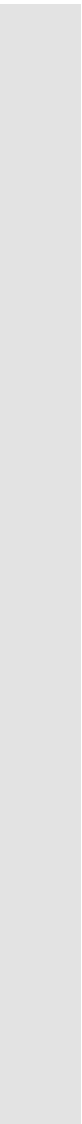
Policies

☒ Advanced filters

Category: **Select risk category**

Hide filters Table settings

Policy templates



DEMO 5

Policies

Deploying Microsoft Defender for Cloud Apps requires the following tasks:

1. **Set instant visibility, protection, and governance actions for your apps (Required).** Connect apps to Microsoft Defender for Cloud Apps.
2. **Protect sensitive information with DLP policies (Recommended).** Enable file monitoring and create file policies.
3. **Control cloud apps with policies (Required).** Policies enable organizations to create governance actions and set data loss prevention and file-sharing controls.
4. **Set up Cloud Discovery (Required).** Enables Microsoft Defender for Cloud Apps to view your cloud app use.
5. **Deploy Conditional Access App Control for catalog apps (Recommended).** Access and session controls in Microsoft Defender for Cloud Apps work with both custom applications and apps from the Cloud app catalog.
6. **Personalize your experience (Recommended).** Customize email settings, set admin notifications, and customize the score metrics.
7. **Organize the data according to your needs (Recommended).** Create IP address tags and continuous reports and add domains for business units.

Deploy Microsoft Defender for Cloud Apps

Troubleshoot Cloud Discovery in Microsoft Defender for Cloud Apps

Microsoft Defender for Endpoint integration

Log parsing errors

Log collector errors

Discovery dashboard errors

Used resources:

<https://learn.microsoft.com/en-us/defender-cloud-apps/>

<https://techcommunity.microsoft.com/t5/security-compliance-and-identity/announcing-microsoft-defender-for-cloud-apps/ba-p/2835842/>

<https://learn.microsoft.com/en-us/defender-cloud-apps/proxy-intro-aad/>

<https://learn.microsoft.com/en-us/defender-cloud-apps/access-policy-aad/>

<https://learn.microsoft.com/en-us/defender-cloud-apps/session-policy-aad/>

<https://learn.microsoft.com/en-us/defender-cloud-apps/proxy-deployment-aad/>

<https://learn.microsoft.com/en-us/defender-cloud-apps/proxy-deployment-any-app/>

<https://www.aka.ms/mcaslicensing/>

<https://learn.microsoft.com/en-us/defender-cloud-apps/app-governance-get-started/>

Q & A

Email: info@zalnet.pl

Twitter: beatazalewa

Github: beatazalewa

Slides:

<https://github.com/beatazalewa/AzureSummit2023>

Thank you for your precious time😊